ບົດທີ 40 ສັງຄະນິດມີຂອບ (ຕໍ່)

ເມື່ອຮຽນຈົບບົດນີ້ນັກຮຽນຈະສາມາດ ເຂົ້າໃຈການຄິດໄລ່ສັງຄະນິດມີຂອບເມື່ອໃຊ້ຕົວປ່ຽນໃໝ່, ເມື່ອໃຊ້ສູດສັງຄະນິດພາກສ່ວນ ຫຼື ເມື່ອຂອບແມ່ນອະສົງໄຂ.

ການຄິດໄລ່ສັງຄະນິດມີຂອບແມ່ນມີຄວາມສຳຄັນຫຼາຍ ຊຶ່ງຊ່ວຍໃຫ້ເຮົາສາມາດຄິດໄລ່ເນື້ອທີ່ເສັ້ນໂຄ້ງ, ບໍລິມາດຂອງຮູບສາມມິຕິ, ຄວາມຍາວຂອງເສັ້ນໂຄ້ງ ແລະ ມວນສານຂອງວັດຖຸທີ່ບໍ່ສະໝ່ຳສະເໝີໄດ້.

1. ສັງຄະນິດມີຂອບ

ຄຳແນະນຳ: ການຄິດໄລ່ສັງຄະນິດມີຂອບ ເມື່ອຕ້ອງໃຊ້ຕົວປ່ຽນໃໝ່, ໃຊ້ສັງຄະນິດພາກສ່ວນ ມີວິທີດັ່ງນີ້:

ວິທີ 1: ແກ້ສັງຄະນິດແບບບໍ່ມີຂອບກ່ອນ ພາຍຫຼັງໄດ້ຄຳຕອບແລ້ວ ຈຶ່ງຄິດໄລ່ສັງຄະນິດມີຂອບ ໂດຍປະຕິບັດຕາມບົດຮຽນທີ່ຜ່ານມາ.

ວິທີ 2: ວາງຕົວປ່ຽນໃໝ່ ພາຍຫຼັງວາງຕົວປ່ຽນໃໝ່ແລ້ວ ໃຫ້ຊອກຂອບຕາມຕົວປ່ຽນໃໝ່ນັ້ນ ແລ້ວ ຈຶ່ງຄິດໄລ່ຕາມບົດຮຽນທີ່ຜ່ານມາ.

ໝາຍເຫດ: ນັກຮຽນສາມາດສຶກສາບົດຕົວຢ່າງຂ້າງລຸ່ມນີ້ເພີ່ມເຕີມ.

ຄຳຖາມ. ຈົ່ງຄິດໄລ່ສັງຄະນິດຕໍ່ໄປນີ້:

1) $\int_0^1 xe^{x^2}\,dx$

2) $\int_0^{\frac{\pi}{2}} \frac{\cos x}{1+\sin x}\,dx$

3) $\int_0^1 \tan^{-1}\,dx$

4) $\int_1^{+\infty} \frac{1}{x^2}\,dx$

5) $\int_{-\infty}^0 xe^x\,dx$

6) $\int_{-\infty}^{+\infty} \frac{1}{1+x^2}\,dx$

2. ສະຫຼຸບ

- ການຄິດໄລ່ສັງຄະນິດມີຂອບເມື່ອໃຊ້ຕົວປ່ຽນໃໝ່ ອາດໃຊ້ສູດ:

$$\int_a^b f\left[g(x)\right]g'(x)dx \overset{u=g(x)}{=} \int_{g(a)}^{g(b)} f(u)du$$

- ການຄິດໄລ່ສັງຄະນິດມີຂອບໂດຍໃຊ້ສູດສັງຄະນິດພາກສ່ວນອາດໃຊ້ສູດ:

$$\int_a^b f(x)g'(x)dx = f(x)g(x)\Big|_a^b - \int_a^b f'(x)g(x)dx$$

- ການຄິດໄລ່ສັງຄະນິດມີຂອບອະສິງໄຂແມ່ນໃຊ້ສູດດັ່ງນີ້:

$$\int_a^{+\infty} f(x)dx = \lim_{b\to+\infty}\int_a^b f(x)dx, \quad \int_{-\infty}^b f(x)dx = \lim_{a\to-\infty}\int_a^b f(x)dx$$

$$\int_{-\infty}^{+\infty} f(x)dx = \int_{-\infty}^c f(x)dx + \int_c^{+\infty} f(x)dx = \lim_{a\to-\infty}\int_a^c f(x)dx + \lim_{b\to+\infty}\int_c^b f(x)dx$$

ຕົວຢ່າງ: ການແກ້ສັງຄະນິດມີຂອບ

1) $\int_0^1 x\left(x^2+1\right)^3 dx$

ວິທີ 1: ຊອກ $\int x\left(x^2+1\right)^3 dx$

ວ່າງ $u = x^2+1$

$$\frac{du}{2} = xdx$$

$$\int x\left(x^2+1\right)^3 dx = \int u^3 \frac{du}{2}$$

$$= \frac{u^4}{8} + C$$

$$= \frac{\left(x^2+1\right)^4}{8} + C$$

ເຮົາໄດ້ $\int_0^1 x\left(x^2+1\right)^3 dx = \left.\frac{\left(x^2+1\right)^4}{8}\right|_0^1$

$$= \left(\frac{\left(1^2+1\right)^4}{8}\right) - \left(\frac{\left(0^2+1\right)^4}{8}\right)$$

$$= \frac{15}{8}$$

ດັ່ງນັ້ນ, $\int_0^1 x\left(x^2+1\right)^3 dx = \frac{15}{8}$

ວິທີ 2: ວາງ $u = x^2+1$

$$\frac{du}{2} = xdx$$

ເມື່ອ $x=0$ ເຮົາໄດ້ $u = 0^2+1 = 1$

ເມື່ອ $x=1$ ເຮົາໄດ້ $u = 1^2+1 = 2$

$$\int_0^1 x\left(x^2+1\right)^3 dx = \int_1^2 u^3 \frac{du}{2}$$

$$= \left(\frac{u^4}{8}\right)\Big|_1^2$$

$$= \left(\frac{2^4}{8}\right) - \left(\frac{1^4}{8}\right)$$

$$= \frac{15}{8}$$

ດັ່ງນັ້ນ, $\int_0^1 x\left(x^2+1\right)^3 dx = \frac{15}{8}$

2) $\int_0^{\frac{\pi}{2}} \cos x \sin^3 x dx$

ວິທີ1: ຊອກ $\int \cos x \sin^3 x dx$

$$\int \cos x \sin^3 x dx = \int \sin^3 x d\left(\sin x\right)$$

$$= \frac{\sin^4 x}{4} + C$$

ເຮົາໄດ້ $\int_0^{\frac{\pi}{2}} \cos x \sin^3 x dx = \left(\frac{\sin^4 x}{4} \right) \Big|_0^{\frac{\pi}{2}}$

$$= \frac{\sin^4 \frac{\pi}{2}}{4} - \frac{\sin^4 0}{4}$$

$$= \frac{1}{4} - \frac{0}{4}$$

$$= \frac{1}{4}$$

ດັ່ງນັ້ນ, $\int_0^{\frac{\pi}{2}} \cos x \sin^3 x dx = \frac{1}{4}$

ວິທີ 2: ວ່າ $u = \sin x,\ du = \cos x dx$

ເມື່ອ $x = 0$ ເຮົາໄດ້ $u = \sin 0 = 0$

ເມື່ອ $x = \frac{\pi}{2}$ ເຮົາໄດ້ $u = \sin \frac{\pi}{2} = 1$

$$\int_0^{\frac{\pi}{2}} \cos x \sin^3 x dx = \int_0^1 u^3 du$$

$$= \left(\frac{u^4}{4} \right) \Big|_0^1$$

$$= \left(\frac{1^4}{4} \right) - \left(\frac{0^4}{4} \right)$$

$$= \frac{1}{4}$$

ດັ່ງນັ້ນ, $\int_0^{\frac{\pi}{2}} \cos x \sin^3 x dx = \frac{1}{4}$

3) $\int_0^{\frac{\pi}{2}} x \sin x dx$

ຊອກ $\int x \sin x dx$

ວ່າ $u = x$

$dv = \sin x dx$

ຈະໄດ້ $du = dx$

$v = -\cos x$

ນຳໃຊ້ສູດ $\int udv = uv - \int vdu$

$\int x\sin xdx = -x\cos x - \int(-\cos x)dx$

$\int x\sin xdx = -x\cos x + \sin x + C$

ເຮົາໄດ້ $\int_0^{\frac{\pi}{2}} x\sin xdx = \left(-x\cos x + \sin x\right)\Big|_0^{\frac{\pi}{2}}$

$= \left(-\frac{\pi}{2}\cos\frac{\pi}{2} + \sin\frac{\pi}{2}\right) - \left(-0\cdot\cos 0 + \sin 0\right)$

$= 1 - 0$

$= 1$

ດັ່ງນັ້ນ, $\int_0^{\frac{\pi}{2}} x\sin xdx = 1$

4) $\int_0^{\frac{\pi}{2}} \frac{\cos x}{1+\sin x}dx$

ວາງ $u = 1+\sin x$

$du = \cos xdx$

ເມື່ອ $x = 0$ ເຮົາໄດ້ $u = 1+\sin 0 = 1$

ເມື່ອ $x = \frac{\pi}{2}$ ເຮົາໄດ້ $u = 1+\sin\frac{\pi}{2} = 2$

$\int_0^{\frac{\pi}{2}} \frac{\cos x}{1+\sin x}dx = \int_1^2 \frac{1}{u}du$

$= \left(\ln u\right)\Big|_1^2$

$= \ln 2 - \ln 1$

$= \ln 2$

ດັ່ງນັ້ນ, $\int_0^{\frac{\pi}{2}} \frac{\cos x}{1+\sin x}dx = \ln 2$

5) $\int_{\frac{\sqrt{2}}{2}}^{1} \sin^{-1} x\,dx$

ຊອກ $\int \sin^{-1} x dx$

ຈາກຕົວຢ່າງໃນບົດຮຽນບົດທີ 38 ຜ່ານມາ ເຮົາມີ:

$$\int \sin^{-1} x dx = x\sin^{-1} x + \sqrt{1-x^2} + c$$

ເຮົາໄດ້ $\int_{\frac{\sqrt{2}}{2}}^{1} \sin^{-1} x\,dx = \left(x\sin^{-1} x + \sqrt{1-x^2}\right)\Big|_{\frac{\sqrt{2}}{2}}^{1}$

$$= \left(1.\sin^{-1} 1 + \sqrt{1-1^2}\right) - \left(\frac{\sqrt{2}}{2}\sin^{-1}\frac{\sqrt{2}}{2} + \sqrt{1-\left(\frac{\sqrt{2}}{2}\right)^2}\right)$$

$$= \left(\frac{\pi}{2}\right) - \left(\frac{\sqrt{2}}{2}.\frac{\pi}{4} + \frac{\sqrt{2}}{2}\right)$$

$$= \frac{\pi\left(4-\sqrt{2}\right)}{8} - \frac{\sqrt{2}}{2}$$

ດັ່ງນັ້ນ, $\int_{\frac{\sqrt{2}}{2}}^{1} \sin^{-1} x\,dx = \frac{\pi\left(4-\sqrt{2}\right)}{8} - \frac{\sqrt{2}}{2}$

6) $\int_{-\infty}^{+\infty} xe^{-x^2} dx$

ຊອກ $\int xe^{-x^2} dx$

ວ່າ $u = -x^2$

ເຮົາໄດ້ $\frac{du}{dx} = -2x$

$$\frac{du}{-2} = xdx$$

$$\int xe^{-x^2} dx = -\frac{1}{2}\int e^u du = -\frac{1}{2}e^u + C$$

$$\int xe^{-x^2} dx = -\frac{1}{2}e^{-x^2} + C$$

ເຮົາໄດ້: $\int_{-\infty}^{+\infty} xe^{-x^2}dx = \int_{-\infty}^{0} xe^{-x^2}dx + \int_{0}^{+\infty} xe^{-x^2}dx$

$$\int_{-\infty}^{+\infty} xe^{-x^2}dx = \lim_{a\to-\infty}\int_{a}^{0} xe^{-x^2}dx + \lim_{b\to+\infty}\int_{0}^{b} xe^{-x^2}dx$$

$$= \lim_{a\to-\infty}\left(-\frac{1}{2}e^{-x^2}\right)\Big|_{a}^{0} + \lim_{b\to+\infty}\left(-\frac{1}{2}e^{-x^2}\right)\Big|_{0}^{b}$$

$$= \lim_{a\to-\infty}\left(-\frac{1}{2}e^{0} - \left(-\frac{1}{2}e^{-a^2}\right)\right) + \lim_{b\to+\infty}\left(-\frac{1}{2}e^{-b^2} - \left(-\frac{1}{2}e^{0}\right)\right)$$

$$= \lim_{a\to-\infty}\left(-\frac{1}{2} + \frac{1}{2}e^{-a^2}\right) + \lim_{b\to+\infty}\left(-\frac{1}{2}e^{-b^2} + \frac{1}{2}\right)$$

$$= \left(-\frac{1}{2} + 0\right) + \left(0 + \frac{1}{2}\right)$$

$$= 0$$

ດັ່ງນັ້ນ, $\int_{-\infty}^{+\infty} xe^{-x^2}dx = 0$

ບົດຝຶກຫັດ

ຈົ່ງຄິດໄລ່ສົງຄະນິດຕໍ່ໄປນີ້:

1. $\int_{0}^{1} \frac{x^2}{3-x^3}dx$
2. $\int_{1}^{2} \frac{2x}{\left(x^2+1\right)^2}dx$
3. $\int_{0}^{1} x\sqrt{x^2+3}dx$
4. $\int_{1}^{2} \frac{dx}{\left(3-5x\right)^2}$
5. $\int_{0}^{\frac{\pi}{3}} \frac{\sin x}{\cos^2 x}dx$
6. $\int_{1}^{4} \ln\sqrt{x}\,dx$
7. $\int_{e}^{+\infty} \frac{1}{x\left(\ln x\right)^2}dx$
8. $\int_{-\infty}^{0} e^{3x}dx$
9. $\int_{-\infty}^{+\infty} xe^{-x^2}dx$
10. $\int_{0}^{\frac{1}{2}} \cos^{-1} 2x\,dx$
11. $\int_{0}^{1} x\tan^{-1} xdx$
12. $\int_{0}^{\frac{1}{2}} \sin^{-1} x\,dx$
13. $\int_{0}^{\frac{\sqrt{3}}{3}} \tan^{-1} 3x\,dx$
14. $\int_{0}^{\frac{1}{2}} \sin^{-1} 2x\,dx$
15. $\int_{0}^{1} x^2\tan^{-1} xdx$